Assessment Menu: Teacher Guide

The Shaded Region Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
BOUNDARY	Graph the boundary line (solid or dashed)	A1, A3, E1, E6
SHADE	Test points and shade the solution	A2, A4, E2, E4, S3
WRITE	Write the inequality from a description	A5, E3, S1, S2
MODEL	Use inequalities in real-world situations	A6, E5, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 BOUNDARY	Should the boundary line be solid or dashed? **(a)** $y \geq x - 3$ **(b)** $y < 2x + 1$	**Answer:** (a) solid (b) dashed (a) "≥" includes points on the line, so the line is **solid**. (b) "<" does not include points on the line, so the line is **dashed**.	2 pts: (a) solid 2 pts: (b) dashed 1 pts: Reason: ≥ or ≤ includes the line, < or > does not
A2 SHADE	Test the point (0, 0) in $y < 2x + 3$. Is it a solution? Should you shade the side of the line that contains the origin?	**Answer:** Yes: $0 < 3$ is true, so shade the side containing the origin. Substitute $x = 0$, $y = 0$: $0 < 2(0) + 3$, so $0 < 3$. This is true, so (0, 0) is a solution and its side of the line is shaded.	2 pts: Substitutes (0, 0) correctly 1 pts: Decides $0 < 3$ is true 2 pts: Shade the side containing the origin
A3 BOUNDARY	The boundary of $y \leq 2x - 1$ is the line $y = 2x - 1$. Find y when $x = 0$ and when $x = 2$, and say whether the line is solid or dashed.	**Answer:** (0, −1) and (2, 3); the line is solid. $x = 0$: $y = 2(0) - 1 = -1$, so the point (0, −1). $x = 2$: $y = 2(2) - 1 = 3$, so the point (2, 3). "≤" includes the line, so it is **solid**.	1 pts: $y = -1$ when $x = 0$ 1 pts: $y = 3$ when $x = 2$ 1 pts: Both points written as ordered pairs 2 pts: Solid, with a reason (≤ includes the line)
A4 SHADE	Which of the points (2, 3), (1, 1), (0, 0) is a solution of $y > 3x - 2$? Show the test for each.	**Answer:** Only (0, 0). (2, 3): $3 > 3(2) - 2 = 4$? False. (1, 1): $1 > 3(1) - 2 = 1$? False (1 is not greater than 1; the point is on the line). (0, 0): $0 > -2$? True.	3 pts: Tests all three points by substituting 2 pts: Only (0, 0), with (1, 1) rejected because it lies on the dashed line
A5 WRITE	A **dashed** line passes through (0, −2) and has slope 3. The region **above** the line is shaded. Write the inequality.	**Answer:** $y > 3x - 2$ The line is $y = 3x - 2$. Dashed means > or <; "above" means y is greater: $y > 3x - 2$.	2 pts: Boundary equation $y = 3x - 2$ 1 pts: Dashed line gives a strict sign 2 pts: Above gives >: $y > 3x - 2$
A6 MODEL	Maya has at most $30 for notebooks ($3 each) and pens ($2 each). Write an inequality for *n* notebooks and *p* pens. Is (6, 6) a solution?	**Answer:** $3n + 2p \leq 30$. Yes: $3(6) + 2(6) = 30$. Total cost is $3n + 2p$ and it must be at most 30: $3n + 2p \leq 30$ Test (6, 6): $18 + 12 = 30 \leq 30$ is true, so yes (she spends exactly $30).	2 pts: Correct costs $3n + 2p$ 1 pts: Uses ≤ (at most) 2 pts: Tests (6, 6) and answers yes

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 BOUNDARY	The boundary of $2x - 3y > 6$ is the line $2x - 3y = 6$. **(a)** Find its x- and y-intercepts. **(b)** Use them to find its slope. **(c)** Is the line solid or dashed? Why?	**Answer:** (a) (3, 0) and (0, −2) (b) $\frac{2}{3}$ (c) dashed, because the sign is $>$ (a) $y = 0$: $2x = 6$, so $x = 3$. $x = 0$: $-3y = 6$, so $y = -2$. (b) Slope $= \frac{-2 - 0}{0 - 3} = \frac{2}{3}$ (c) The inequality uses "$>$", so points on the line are not solutions: **dashed**.	2 pts: x-intercept (3, 0) 2 pts: y-intercept (0, −2) 3 pts: Slope $\frac{2}{3}$ from the two intercepts 3 pts: Dashed, with a reason
E2 SHADE	For $y \le -\frac{1}{2}x + 2$, test the points (0, 0), (4, 2), and (−2, 3). Which are solutions? Then say whether you shade above or below the solid line.	**Answer:** (0, 0), (−2, 3) yes; (4, 2) no. Shade below. (0, 0): $0 \le 2$ ✓. (4, 2): $2 \le -2 + 2 = 0$? False. (−2, 3): $3 \le 1 + 2 = 3$ ✓ (it is on the solid line). The solution points lie below or on the line (the origin is below it), so shade **below**.	2 pts: Tests (0, 0) correctly 2 pts: Tests (4, 2) correctly: not a solution 3 pts: Tests (−2, 3): a solution, on the solid line 3 pts: Shade below the line
E3 WRITE	A **solid** line passes through (−2, 1) and (2, 5). The region **below** the line is shaded. Write the inequality in slope-intercept form.	**Answer:** $y \le x + 3$ Slope $= \frac{5 - 1}{2 - (-2)} = \frac{4}{4} = 1$ Use (2, 5): $b = 3$, so $y = x + 3$. Solid and below: $y \le x + 3$.	3 pts: Correct slope 1 2 pts: Boundary $y = x + 3$ 2 pts: Solid gives $\le$ 3 pts: Below gives $\le$: $y \le x + 3$
E4 SHADE	Dana graphed $x - 2y < 4$: $-2y < -x + 4$ $y < \frac{1}{2}x - 2$, so she shaded **below** the dashed line. Find her mistake, fix it, and confirm with a test point.	**Answer:** She did not flip the sign when dividing by −2. It should be $y > ½x - 2$ (shade above); (0, 0) confirms it. Dividing by −2 reverses the sign: $y > \frac{1}{2}x - 2$ (shade **above**). Test (0, 0): $0 < 4$ is true, and the origin is above the line (it passes through (0, −2)).	4 pts: Identifies the missing sign flip 2 pts: Explains why a negative divisor reverses the inequality 2 pts: Correct inequality and shading above 2 pts: Test point (0, 0) used to confirm
E5 MODEL	A club sells popcorn for $3 and drinks for $2. It needs to raise **at least $120**. Write an inequality for *p* popcorns and *d* drinks. Give two combinations that work and one that does not.	**Answer:** $3p + 2d \ge 120$. Works: (30, 15) and (40, 10). Does not work: (20, 20). Sales are $3p + 2d$ and must be at least 120: $3p + 2d \ge 120$ (30, 15): 120 ✓ (40, 10): 140 ✓ (20, 20): 100 < 120, fails.	3 pts: Correct inequality ($\ge$, with $3p + 2d$) 3 pts: Two combinations that check 2 pts: One combination that fails 2 pts: Shows the substitution for each
E6 BOUNDARY	Solve $-2y > 6$. Describe the boundary line (type, equation, solid or dashed) and the shading.	**Answer:** $y < -3$: dashed horizontal line, shade below. Divide by −2 and **flip** the sign: $y < -3$. Boundary $y = -3$: **horizontal**, and **dashed** (strict sign). Shade below it.	3 pts: $y < -3$ (flips) 3 pts: Horizontal, dashed 4 pts: Shades below

Answer Key: Chef's Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 WRITE	A **dashed** line passes through (0, 5) and (4, 2). The origin lies in the shaded region. **(a)** Write the inequality in slope-intercept form. **(b)** Rewrite it with whole-number coefficients (no fractions). **(c)** Is (4, 2) a solution? Is (4, 1)?	**Answer:** (a) $y < -\frac{3}{4}x + 5$ (b) $3x + 4y < 20$ (c) (4, 2): no; (4, 1): yes (a) Slope = $\frac{2-5}{4-0} = -\frac{3}{4}$; line $y = -\frac{3}{4}x + 5$. The origin is shaded: $0 < 5$, so the region is below. Dashed, so $y < -\frac{3}{4}x + 5$. (b) Multiply by 4: $4y < -3x + 20$, so $3x + 4y < 20$. (c) (4, 2): $12 + 8 = 20$, and $20 < 20$ is false (on the dashed line), so no. (4, 1): $12 + 4 = 16 < 20$, so yes.	4 pts: Slope $-\frac{3}{4}$ 4 pts: Dashed, below: < 4 pts: (b) $3x + 4y < 20$ 4 pts: (4, 2): no (on dashed line) 4 pts: (4, 1): yes ($16 < 20$)
S2 WRITE	A system's solution region lies **on or below** a solid line through (0, 6) and (3, 0), and **above** a dashed line through (0, −2) and (2, 0). **(a)** Write the system of inequalities. **(b)** Is (1, 1) in the solution region? **(c)** Is (2, 0)?	**Answer:** (a) $y \le -2x + 6$ and $y > x - 2$ (b) Yes (c) No (a) Solid: slope −2, so $y \le -2x + 6$. Dashed: slope 1, so $y > x - 2$. (b) (1, 1): $1 \le -2 + 6 = 4$ ✓ and $1 > 1 - 2 = -1$ ✓, so yes. (c) (2, 0): $0 \le 2$ ✓ but $0 > 2 - 2 = 0$ is false (on the dashed line), so no.	4 pts: $y \le -2x + 6$ 4 pts: $y > x - 2$ 4 pts: (b) Tests (1, 1): yes 4 pts: (c) Tests (2, 0) 4 pts: (c) No: on the dashed line
S3 SHADE	Graph the system on the grid: $y > 2x - 3$ and $x + y \le 6$. Shade only the region that satisfies both, and label the point where the boundaries cross. Is that point a solution? Explain.	**Answer:** Dashed $y = 2x - 3$ and solid $x + y = 6$ cross at (3, 3). The solution region is the overlap containing the origin. (3, 3) is not a solution (on the dashed line). $y = 2x - 3$ through (0, −3), (3, 3): **dashed**. $x + y = 6$ through (0, 6), (6, 0): **solid**. Crossing: $2x - 3 = 6 - x$, so $x = 3$, $y = 3$. Test (0, 0): $0 > -3$ ✓ and $0 \le 6$ ✓: shade the overlap containing the origin. (3, 3): $3 > 3$ is false (on the dashed line), so it is not a solution.	4 pts: Dashed line $y = 2x - 3$ 4 pts: Solid line $x + y = 6$ 4 pts: Overlap shaded 4 pts: Crossing point (3, 3) labeled 4 pts: (3, 3) not a solution: on dashed line
S4 MODEL	Alex works at most 10 hours a week tutoring (*t* hours, $12 an hour) and dog-walking (*d* hours, $8 an hour). He wants **at least $96**. **(a)** Write a system (include $t \ge 0$, $d \ge 0$). **(b)** Graph and shade it (*t* across, *d* up). **(c)** Do 5 tutoring and 4 dog-walking hours work?	**Answer:** (a) $t + d \le 10$, $12t + 8d \ge 96$, $t \ge 0$, $d \ge 0$ (b) triangle (4, 6), (8, 0), (10, 0) (c) No: he earns only $92. (a) $t + d \le 10$; $12t + 8d \ge 96$ (or $3t + 2d \ge 24$); $t \ge 0$; $d \ge 0$. (b) Solid lines $t + d = 10$ and $3t + 2d = 24$ cross at (4, 6). Shade below the first, above the second, in quadrant I: triangle (4, 6), (8, 0), (10, 0). (c) (5, 4): $12(5) + 8(4) = 92 < 96$. No.	5 pts: $t + d \le 10$, $t \ge 0$, $d \ge 0$ 3 pts: $12t + 8d \ge 96$ 4 pts: Both solid lines graphed 4 pts: Correct region shaded (triangle) 4 pts: (c) Tests (5, 4): earns only $92